

Security Onion Home Lab

Complete Setup & Configuration Documentation

Bryan Sykes | March 2026

This document covers the full end-to-end installation and configuration of Security Onion 2.4 in EVAL mode on Oracle VirtualBox running on Windows, including all troubleshooting steps encountered and resolved during the process.

Lab Environment

Component	Details
Host OS	Windows 11
Virtualization	Oracle VirtualBox
Security Onion Version	2.4.211
Base OS	Oracle Linux Server 9.7
Kernel	5.15.0-318.199.3.2.el9uek.x86_64
Installation Mode	EVAL (Standalone)
Admin Username	soadmin
Final Management IP	192.168.1.196
Network Mode	Bridged Adapter (Realtek PCIe GbE)
Attacker VM	Kali Linux (VirtualBox)

Section 1: Pre-Installation Requirements

1.1 Minimum Hardware Requirements

Security Onion EVAL mode is resource-intensive due to its full ELK stack (Elasticsearch, Logstash, Kibana) plus Suricata and Zeek running as Docker containers.

Resource	Minimum	Recommended	Notes
RAM	8 GB	12–16 GB	Allocate to VM, not total system
Disk Space	200 GB	250 GB+	SO stores full PCAPs — storage fills fast
CPU Cores	4 cores	4–6 cores	Allocate in VirtualBox settings
Network Adapters	2 NICs	3 NICs	Mgmt + Monitor + Host access

***WARNING:** Allocating less than 200 GB of disk space WILL cause the installation to fail. This was confirmed during this setup — the first install attempt used 100 GB and failed at the final configuration stage.*

1.2 Software Downloads

- Oracle VirtualBox — <https://www.virtualbox.org>
- Security Onion ISO — <https://securityonionsolutions.com/software>
- Kali Linux ISO (for attacker VM) — <https://www.kali.org/get-kali/>

Section 2: Creating the VirtualBox VM

2.1 Creating the Virtual Machine

Open VirtualBox and click New to create a new virtual machine.

Setting	Value
Name	SECURITYONION (or any preferred name)
Type	Linux
Version	Other Linux (64-bit)
RAM	8192 MB minimum (12288 MB recommended)

CPU Cores	4
Disk Type	VDI (VirtualBox Disk Image)
Disk Size	200 GB minimum — CRITICAL
Disk Allocation	Dynamically Allocated

WARNING: On the guided installation screen, check the 'Skip Unattended Installation' box at the bottom before proceeding. Failure to do this caused the first install attempt to fail with VERR_ALREADY_EXISTS errors related to auto-generated kickstart files conflicting with Security Onion's own installer.

2.2 Network Adapter Configuration

Security Onion requires multiple network adapters. Configure them in Settings > Network:

Adapter	Type	Purpose
Adapter 1	Bridged Adapter (Realtek PCIe GbE Family Controller)	Management — gives SO a real LAN IP accessible from Windows browser
Adapter 2	Internal Network (name: labnet)	Monitor/Sniffing — captures attack traffic from Kali VM
Adapter 3	Host-Only (optional)	Originally attempted for web UI access before switching to Bridged

NOTE: Initially Adapter 1 was set to NAT, which caused extensive web UI access issues. The correct final configuration uses Bridged mode pointed at the physical Ethernet controller. NAT assigns an internal 10.0.2.x address that is not reachable from the Windows host browser.

2.3 Mounting the ISO

In Settings > Storage, click the empty optical drive, then select the Security Onion .iso file. Confirm it appears under the IDE controller before booting.

Section 3: Running the Security Onion Installer

3.1 Initial Boot and Disk Warning

Boot the VM. Security Onion will load a text-based installer and display a warning screen:

```
** WARNING **
Installing the Security Onion ISO on this device will DESTROY ALL DATA
```

```
and partitions! ALL DATA WILL BE LOST.  
Do you wish to continue? (Type the entire word 'yes' to proceed.)
```

Type the full word yes and press Enter to proceed. This only affects the virtual disk — no data on the Windows host is at risk.

3.2 Installer Progress

The installer runs automatically after confirmation. It performs the following operations:

- Partitioning the virtual disk (/dev/sda)
- Creating logical volumes: system-nsm, system-root, system-tmp, system-swap
- Formatting with XFS filesystem
- Installing Oracle Linux base packages
- Setting admin username and password

NOTE: This stage takes 20–40 minutes depending on system performance. The screen may appear to hang — this is normal. Do not reset or power off the VM during this phase.

Section 4: Security Onion Setup Wizard (EVAL Mode)

After the base OS installs and the VM reboots, log in with the admin credentials set during installation. Security Onion launches a blue-screen text-based configuration wizard.

4.1 Setup Wizard Step-by-Step

Step	Screen	Selection Made	Notes
1	Install Type	EVAL	Best mode for home lab — includes all components on one node
2	Terms Agreement	Yes/Agree	Accept to continue
3	Hostname	securityoniontester	Keep simple, no spaces or special characters

4	Management NIC	enp0s3	First adapter — Bridged/management interface
5	Connection Type	Direct	No proxy in home lab environment
6	Monitor NIC	enp0s8	Second adapter — sniffing/monitoring interface, selected with spacebar
7	Web UI Access	IP	Use IP address to access web interface
8	CIDR/Allowed IPs	192.168.x.x/32	Windows host IP with /32 CIDR — only allow your machine
9	SOC Admin Email	admin@lab.local	Email used to log into web dashboard
10	SOC Admin Password	[set during wizard]	Used to log into https://[management-ip]
11	Final Confirmation	Yes	Review summary and confirm to begin backend setup

NOTE: The wizard assigns Security Onion's management IP based on whatever address the management NIC holds at setup time. If the adapter type changes later (e.g. NAT to Bridged), the IP will change and SO's internal config must be updated using the so-ip-update command.

4.2 Post-Wizard Background Setup

After confirmation, Security Onion runs a long background configuration process that:

- Pulls and configures all Docker containers
- Sets up Suricata IDS engine
- Sets up Zeek network analysis framework
- Configures Elasticsearch and Kibana
- Initializes the SOC web dashboard (so-soc, so-nginx)
- Applies firewall rules and network configuration

NOTE: This process takes 20–45 minutes and runs silently with a black screen. This is expected behavior — do not interrupt it.

Section 5: Troubleshooting Log

The following issues were encountered and resolved during this installation. Each is documented as a learning reference.

Issue 1: VERR_ALREADY_EXISTS on First Boot

Field	Details
Error	VISO: Error setting name '/ks.cfg' — VERR_ALREADY_EXISTS
Cause	VirtualBox's Unattended Install feature auto-generated conflicting kickstart and .viso files
Resolution	Deleted Unattended-*.cfg and Unattended-*.viso files from the VM folder. Recreated VM with Skip Unattended Installation checked.
Prevention	Always check Skip Unattended Installation when creating VMs for Security Onion

Issue 2: Forgotten Admin Username

Field	Details
Problem	Admin username forgotten after initial OS install
Approach	Booted into single-user mode via GRUB editor
GRUB Edit	Added rw init=/bin/bash to the linux kernel line, then Ctrl+X to boot
Commands Used	mount -o remount,rw / then ls /home to discover username
Username Found	mrbtechguy07
Password Reset	passwd mrbtechguy07 followed by sync then exec /sbin/init
Outcome	Reset succeeded on second attempt after correctly remounting filesystem first

Issue 3: Installation Failed Due to Insufficient Disk Space

Field	Details
Problem	Security Onion setup wizard failed during backend configuration
Root Cause	Virtual disk was only 100 GB — below the 200 GB minimum required

Resolution	Deleted VM entirely and recreated with 200 GB virtual disk
Lesson	Security Onion stores full packet captures (PCAPs) and ELK data — storage requirements are non-negotiable

Issue 4: Web UI Inaccessible — NAT Network Mode

Field	Details
Problem	Browser could not reach Security Onion web UI at https://10.0.2.15
Root Cause	Adapter 1 was set to NAT mode. NAT assigns an internal 10.0.2.x IP that is not routable from the Windows host
Attempted Fixes	Port forwarding rules (8443:443), Host-Only adapter, so-allow, so-firewall commands, iptables flush
Final Resolution	Changed Adapter 1 from NAT to Bridged Adapter (Realtek PCIe GbE Family Controller), then ran sudo so-ip-update to update SO's internal config to the new IP
Result	SO assigned IP 192.168.1.196 on the local network, web UI became accessible

Issue 5: so-ip-update Required After Adapter Change

Field	Details
Problem	After switching to Bridged mode, login screen still showed old IP (10.0.2.15). Browser timed out on new IP (192.168.1.150/196)
Root Cause	Security Onion's internal configuration files were hardcoded to the original NAT IP from setup time
Resolution	Ran sudo so-ip-update — this command updates the management IP across all SO config files and Docker container configurations
After Command	Login message updated to https://192.168.1.196 and web UI became accessible

Section 6: Final Verification

6.1 Confirming All Services Running

After successful setup, run the following command to verify all containers are healthy:

```
sudo so-status
```

Expected output — all containers should show 'running':

Container	Expected Status
so-dockerregistry	running
so-elastalert	running
so-elastic-fleet	running
so-elasticsearch	running
so-influxdb	running
so-kibana	running
so-nginx	running
so-sensoroni	running
so-soc	running
so-suricata	running
so-zeek	running
so-teleggraf	running

SUCCESS: All containers confirmed running. Final status message: 'This onion is ready to make your adversaries cry!'

6.2 Accessing the Web Interface

Open a browser on the Windows host and navigate to:

```
https://192.168.1.196
```

Accept the SSL certificate warning (self-signed cert is expected in a home lab). Log in with:

- Email: the address set during wizard setup
- Password: the SOC admin password set during wizard setup

SUCCESS: Security Onion web dashboard (SOC) confirmed accessible and loading at https://192.168.1.196

Section 7: Next Steps — Portfolio Project

With Security Onion fully operational, the following project will be executed and documented for portfolio and LinkedIn:

Project: Nmap Reconnaissance & SSH Brute Force Detection

Phase	Actions	Tools Used
1 — Setup	Configure Kali VM on labnet Internal Network. Confirm SO can see Kali traffic.	VirtualBox, Kali Linux
2 — Recon	Run Nmap scans from Kali against SO VM: ping sweep, aggressive scan, service version scan, OS detection	Nmap
3 — Attack	Run SSH brute force simulation from Kali using Hydra with rockyou.txt wordlist	Hydra
4 — Detect	Investigate alerts in Security Onion SOC dashboard. Review Suricata IDS alerts.	Security Onion SOC
5 — Analyze	Review Zeek conn.log for port sweep evidence. Pull PCAP and analyze in Wireshark.	Zeek, Wireshark, Kibana
6 — Document	Write formal incident report with IOCs, timeline, evidence, and recommendations.	Word/PDF

Skills Demonstrated by This Project

- Network traffic analysis and packet capture (PCAP)
- Intrusion Detection System (IDS) alert triage — Suricata
- Network protocol analysis — Zeek/Bro
- Log correlation and threat hunting — Kibana
- Incident response documentation
- Attack simulation and adversary emulation
- Linux system administration
- Virtualization and network architecture design

Certifications & Learning Context

This lab project supports active study for CompTIA Security+ and directly applies concepts from:

- NIST RMF — risk identification and incident response
- SOC analyst workflows — alert triage, investigation, escalation
- Network security monitoring — IDS/IPS, PCAP analysis
- Threat detection engineering — signature-based detection with Suricata